Class Test

CO2_AT3

Computer

Vision

Name: S. Amulya
Reg no : 192321044
Date : 21/10/26

Class Test

CO2_AT3

Computer Vision

Name: S. Amulya
Reg no : 192321044
Date : 21/10/26

1. Image Processing

1. Brightening a Dark Image

Log Transformation or Gamma Transformation

Steps:

1. Read the input image.
2. Apply log transformation or gamma correction
3. Normalization the pixel valug
4. Display the enchanced image

Formula: $S = c \log(1+r)$

Result: Dark regions become brighter and details of the museom exhibits become visible

In gamma correction, choosing $\gamma < 1$ increasg the brightneg of dark regions.

2. Fingerprint Enhancement Using Histogram Equalization

It improvg image contrast by spreading pixel intensity values across a wider range. It helps make fingerprint ridgg and valleys clearer.

Processing steps:

- Read the fingerprint image and convert it to grayscale.
- Calculate the histogram of pixel intensity values.
- Calculate the cumulative distribution function.

- Display the enhanced image

formula:

$$s_k = \frac{L-1}{MN} \sum_{j=0}^{k} n_j$$

Here, L is the no. of intensity levels, MN is the total no. of pixels, and n_j is no. of pixels at intensity j.

Result: The fingerprint contrast improves, making ridges and valleys easier to identify.

3 Removing Salt-and-Pepper Noise

The technique we have used Median-Filtering

Salt-and-pepper noise appears as random black and white pixels. A median filter removes these noisy pixels while preserving object edges.

Processing steps:

1. Read the noisy image
2. Select 3x3 neighborhood around each pixel.
3. Arrang the pixel values in ascending order.
4. Find the median value & replace the center pixel with it
5. Repeat the proces & display the filtered image.

Result: The noise is reduced while the important edge remain clear.

4. Sharpening a Circuit Board Image:

Technique we have used Laplacian Sharpening or Unsharp Masking. Image sharpening improves the visibility of edges & fine details in a blurred circuit board image. It helps to identify copper tracks clearly.

Processing steps:

1. Read the circuit board image
2. Convert it into grayscale if necessary
3. Apply a Laplacian filter or create a blurred version of image
4. Extract the detail info & combaine it with the original image

Formula for unsharp masking

$$g = f + k(f - f_{blur})$$

5. Removing Grainy Noise Using a Low-Pass Filter

Technique: Frequency-domain Low-Pass filtering

A low-pass filter reduce high-frequency noise in a drone image & helps reveal the outlines of farm plots.

1. Read the image & convert it to gray scale.

2. Apply the fourier Transformation to convert it into the frequency domain.
3. Apply a low-pass filter such as ideal, Butterworth, or Gaussian.
4. Remove the unwanted high-frequency components.

Effect of cutoff frequency:

- Low cutoff: Removes more noise but may blur edges
- High cutoff: Preserves more details but may leave some noise

6. Enhancing Retinal Blood vessels.

Technique: High-Pass filtering or High-Boost filtering
High-pass filtering enhances fine details & rapid intensity changes in retinal image. It helps make thin blood vessels more visible.

1. Read the retinal fundus image
2. Convert it to gray scale.
3. Apply fourier Transform
4. Use a high-pass filter to emphasize high-frequency details.
5. Apply the inverse Fourier Transform and display

the enhanced image

Formula for high-boost filtering

$$g = Af - f_{blur}, \quad A > 1$$

Result: Blood vessel edges & fine details become mor visible against the background.

7. Detecting Isolated points and Thin lines on Glass Bottles.

Point detection identifies isolated air bubbles, while line detection identifies thin cracks on the bottle surface

1. Read the bottle image and convert it to grayscale.
2. Apply a point detection mask to identify isolated intensity changes.
3. Apply horizontal, vertical, and diagonal line masks to detect cracks.
4. Calculate the filter response for each pixel.
5. Compare the response with a threshold & mark the detected defects

Point detection mask:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Threshold selection: A suitable threshold is choosen based on filter response & background noise. Response above the threshold are marked as possible defects.

8. Separating Handwriting from the Background

Technique: Adapative Thresholding

Adaptive thresholding separates dark handwriting from a white page when the lighting is uneven. It calculates different thresholds for different image regions.

Processing steps:

1. Read the answer sheet image
2. Convert it into grayscale.
3. Apply Gaussian smoothing to reduce noise
4. Divide the image into small neighborhoods.
5. Calculate a local threshold for each neighborhood.
6. Convert the image into a binary image with black handwriting and a white background.

Why adaptive thresholding?

- Handles uneven illumination.
- Preserves handwriting in dark and bright regions
- Performs better than global thresholding when lighting varies.

9. Extracting a Lake Using Region Growing

Technique: Region Growing

Region growing extracts a lake by grouping neighboring pixels with similar intensity values. It separates the lake from the surrounding land.

Processing steps:

1. Read the satellite image
2. Select a seed point inside the lake
3. Examine neighboring pixels.
4. Add pixels that satisfy the intensity similarity criterion.
5. Continue until no more neighboring pixels satisfy the condition.
6. Display the extracted lake region.

Similarity criterion:

$$|I(x,y) - \mu_R| < T$$

Here, $I(x,y)$ is the pixel intensity, μ_R is the region's mean intensity, and T is the threshold.

Seed selection: Choose a point well inside the lake, away from the shoreline.

b. Completing Broken Gear Edges

Technique: Edge linking and Boundary Detection

→ Edge linking connects broken edge segments, while boundary detection identifies the complete outline of the mechanical gear.

Processing steps:

1. Read the gear image and convert it to grayscale
2. Apply an edge detector such as Canny.
3. Use local processing to connect nearby edge segments.
4. Apply the Hough Transform to identify geometric patterns, such as straight lines or circles.
5. Use boundary detection to trace the gear outline.
6. Display the final Image

Result: Broken and discontinuous edges are connected where possible, producing a clearer and more complete gear outline.